

1) A retailer of industrial labels faces constant demand  
 $D = 24000 \frac{\text{units}}{\text{year}}$ ,  $S = \frac{\text{EUR}}{\text{order}}$ ,  $h = 6 \frac{\text{EUR}}{\text{unit} \cdot \text{year}}$

a) Calculate the total cost function

$$TC(Q) = \frac{DS}{Q} + \frac{hQ}{2}$$

$$\Rightarrow \frac{24000 \left( \frac{\text{units}}{\text{year}} \right) \cdot 180 \left( \frac{\text{EUR}}{\text{order}} \right)}{Q \left( \frac{\text{units}}{\text{order}} \right)} + \frac{6Q \frac{\text{EUR}}{\text{unit} \cdot \text{year}}}{2}$$

$$\Rightarrow \frac{4320000 \left( \frac{\text{units} \cdot \text{EUR}}{\text{year} \cdot \text{order}} \right)}{Q \frac{\text{units}}{\text{order}}} + 3Q \left( \frac{\text{EUR}}{\text{year}} \right)$$

$$\Rightarrow \frac{4320000 \left( \frac{\text{EUR}}{\text{year}} \right)}{Q} + 3Q \left( \frac{\text{EUR}}{\text{year}} \right) = TC(Q)$$

b) calculate the EOQ  $Q^*$

$\Rightarrow$  Assume no backlog and instantaneous production

$$Q^* = \sqrt{\frac{2SD}{h}} \quad * \text{ From 14's lecture}$$

$$Q^* = \sqrt{\frac{2(180)(24000)}{6}} = 1200 \text{ units}$$

$$\text{Dimensions} = \sqrt{\frac{\text{units}}{\text{year}} \cdot \frac{\text{EUR}}{\text{order}} \frac{\text{unit} \cdot \text{year}}{\text{EUR}}}$$

$$\frac{\text{EUR}}{\text{order}} = \sqrt{\text{units}^2}$$

$$= \text{EUR} \checkmark$$

$$= \text{units}$$

c) Determine the optimal number of orders per year and the optimal cycle length in working days if the firm operates 300 days per year.

⇒ The optimal number of orders per year is that which fulfills the constant demand.

$$D = 24\,000 \frac{\text{units}}{\text{year}}$$

$$Q^* = 1200 \text{ units}$$

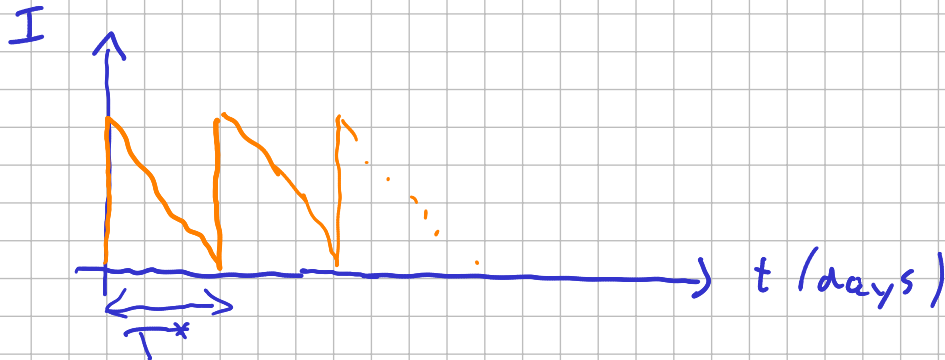
$$\Rightarrow O^* = \frac{D}{Q^*} = \frac{24\,000 \frac{\text{units}}{\text{year}}}{1200 \text{ units}} = 20 \cdot \frac{1}{\text{year}}$$

⇒ The optimal cycle time or period  $T^*$  is the number of days between each order (optimal number of orders  $O^*$ ).

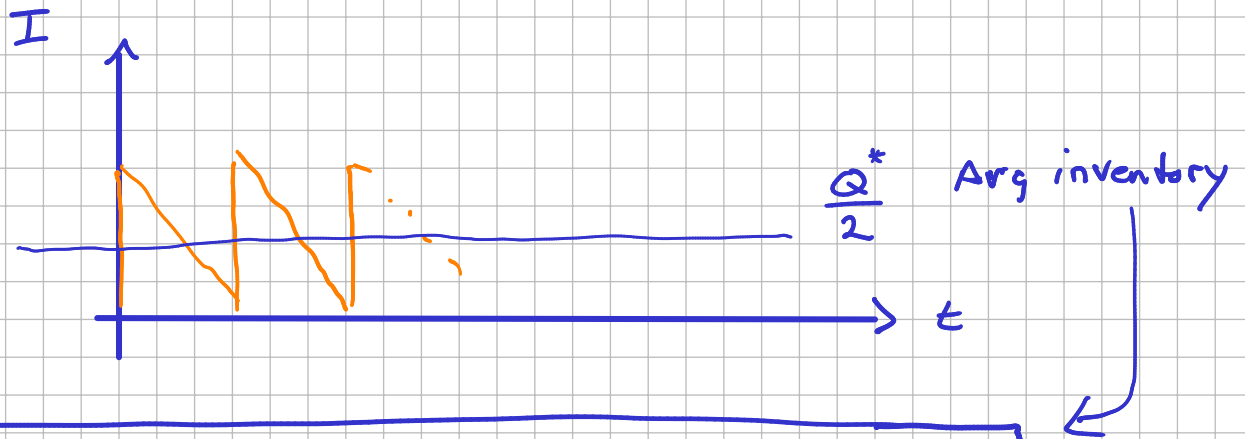
⇒ For 300 working days:

$$T^* = \frac{300 \frac{\text{days}}{\text{year}}}{20 \frac{\text{orders}}{\text{year}}} = \frac{300}{20} \frac{\text{days}}{\text{order}}$$

$$T^* = 15 \frac{\text{days}}{\text{order}}$$



d) Determine the average inventory and the annual ordering cost and holding cost at  $Q^*$ .



$$\Rightarrow \frac{Q^*}{2} = \frac{1200 \text{ units}}{2} = 600 \text{ units}$$

$\Rightarrow$  Ordering cost is how much it costs per year to place all of the orders.

$$Q^* \left( \frac{\text{orders}}{\text{year}} \right) \cdot S \left( \frac{\text{EUR}}{\text{order}} \right)$$

$$\text{Annual ordering Cost} = 20 \cdot 180 \frac{\text{EUR}}{\text{year}} = 3600 \frac{\text{EUR}}{\text{year}}$$

$\Rightarrow$  Annual holding cost is our average inventory multiplied by our holding cost  $h$ .

$$\frac{Q^*}{2} (\text{units}) \cdot h \left( \frac{\text{EUR}}{\text{unit} \cdot \text{year}} \right)$$

$$\text{Annual holding Cost} = 600 \text{ units} \cdot 6 \frac{\text{EUR}}{\text{year}} = 3600 \frac{\text{EUR}}{\text{year}}$$

2] A beverage bottler purchases pallet collars with constant demand.

a) Determine the EOQ  $Q^*$  and the corresponding relevant annual cost:

Given

$$D = 12000, \quad S = 90, \quad h = 3.$$

$\Rightarrow$  Same assumptions as in problem 1

$$\Rightarrow Q^* = \sqrt{\frac{2SD}{h}} = \sqrt{\frac{2(90)(12000)}{3}}$$

$$Q^* = 848.528 \approx 849 \text{ units}$$

H4's  
lecture \*

$$Y_{\min}(Q^*) = \sqrt{2SDh} = \sqrt{2(90)(12000)3}$$

$$Y_{\min}(Q^*) = 2545.584 \frac{\text{EUR}}{\text{year}}$$

b) Compute the total annual cost for  
 $Q = 600$  and  $Q = 1200$ .

$$\Rightarrow TC(Q) = \frac{DS}{Q} + \frac{hQ}{2} \Rightarrow \text{Given}$$

$$\Rightarrow TC(600) = \frac{12000 \cdot 90}{600} + \frac{3 \cdot 600}{2} = 2700 \frac{\text{EUR}}{\text{year}}$$

$$\Rightarrow TC(1200) = \frac{12000 \cdot 90}{1200} + \frac{3 \cdot 1200}{2} = 2700 \frac{\text{EUR}}{\text{year}}$$

$\Rightarrow$  The total cost for both  $Q = 600$  and  $Q = 1200$  are the same!

- At  $Q = 600$ , the ordering cost is high
- At  $Q = 1200$ , the holding cost is high

c) Which of the 2 candidate lot sizes is closer to the optimum in terms of cost?

Quantify the % deviation from the minimum cost in each case.

$\Rightarrow$  Neither is closer - they are both the same!

$$\Rightarrow \% \text{ dev} = \frac{IC - Y_{\min}}{Y_{\min}} \cdot 100 \%$$

$$= \frac{2700 - 2545.584}{2545.584} \cdot 100 \%$$

$\% \text{ dev.} = 6.066 \%$

 $\Rightarrow$  For  $Q=600$  AND  $Q=1200$

d) Determine the average inventory for  $Q=600$ , and  $Q=1200$

$$\frac{Q}{2} \Rightarrow \text{Do the math.}$$

\* This problem illustrates the trade-off. When we have more inventory, we pay for the holding. When we order more frequently, we pay for the ordering costs more often.

[5] A service-parts depot allows planned backorders. The annual demand and cost parameters are given:

$$D = 18\,000, \quad S = 120, \quad h = 4, \quad \pi = 16.$$

a) Write the total annual cost function

$$TC(Q, B) = \underbrace{\frac{DS}{Q}}_{\text{ordering cost}} + \underbrace{\frac{h(Q-B)^2}{2Q}}_{\text{holding cost}} + \underbrace{\frac{\pi B^2}{2Q}}_{\text{Backorder cost}}$$

$$\text{Ordering: } \frac{DS}{Q} = \frac{18\,000 \cdot 120}{Q} = \frac{2\,160\,000}{Q}$$

$$\text{Holding: } \frac{h(Q-B)^2}{2Q} = \frac{4(Q-B)^2}{2Q} = \frac{2(Q-B)^2}{Q}$$

$$\text{Backorder: } \frac{\pi B^2}{2Q} = \frac{16B^2}{2Q} = \frac{8B^2}{Q}$$

Cost Function:

$$TC(Q, B) = \frac{2\,160\,000}{Q} + \frac{2(Q-B)^2}{Q} + \frac{8B^2}{Q}$$

b) Compute the optimal lot size  $Q^*$

$\Rightarrow$  Since we are allowed backorders we use the formula from H4's lecture:

$$Q_2^* = Q_1^* \sqrt{\frac{\pi + h}{h}}$$

$$\Rightarrow Q_1^* = \sqrt{\frac{2SD}{h}} \Rightarrow Q_2^* = \sqrt{\frac{2SD}{h} \left( \frac{\pi + h}{h} \right)}$$

$$Q_2^* = \sqrt{\frac{2 \cdot 18\,000 \cdot 120}{4} \left( \frac{16 + 4}{4} \right)} = 2323.790 \text{ units}$$

c) Compute the optimal minimum backlog  $B^*$  and optimal on-hand inventory

$$I^* = Q^* - B^*$$

$\Rightarrow$  To get  $B^*$ , we derive TC w.r.t  $B$ :

$$\frac{\partial TC(Q, B)}{\partial B} = \frac{\partial}{\partial B} \left[ \frac{DS}{Q} + \frac{h(Q-B)^2}{2Q} + \frac{\pi B^2}{2Q} \right]$$

$$= \frac{\partial}{\partial B} \left[ \frac{h(Q-B)^2}{2Q} + \frac{\pi B^2}{2Q} \right]$$

$$= \frac{h}{Q} (Q-B)(-1) + \frac{2\pi B}{2Q}$$

$$\Rightarrow \frac{\partial TC}{\partial B} = -\frac{h}{Q} (Q-B) + \frac{\pi B}{Q}$$

$\Rightarrow$  Set this to zero to get the 1st-order condition:

$$-\frac{h}{Q} (Q-B) + \frac{\pi B}{Q} = 0$$

$$\frac{h}{Q} (Q-B) = \frac{\pi B}{Q}$$

$$h(Q-B) = \pi B$$

$$hQ - hB = \pi B$$

$$hQ = \pi B + hB$$

$$hQ = (\pi + h)B$$

$$B^* = Q^* \left( \frac{h}{\pi + h} \right)$$

$$B^* = 2323.790 \left( \frac{4}{16+4} \right) = 464.758 \text{ units}$$